Lung Cancer Detection

Detection of Cancer in Human Lung Using Various Deep Learning Techniques

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***Abstract*— In this research a Lung Cancer Detection System is proposed which is fast, accurate and a stable way to detect cancer in lungs. Transfer Learning along with Convolutional Neural Network was used on several Deep Learning Models like ResNet, DenseNet, VGG, MobileNet and EfficientNet to find which model works the best for our Current dataset in order to give the best results. The system was evaluated on Chest CT Scan Dataset by Kaggle. The most efficient trained model produced an accuracy of 96%, Precision of 55% and Recall of 81% after 100 epochs. A confidence interval for the population means of the validation accuracies verified that the proposed system would produce consistent results for multiple trials.**

Keywords- Transfer Learning; Convolutional Neural Networks; ResNet; DenseNet; MobileNet; VGG; MobileNet

I. Introduction

Cancer is a term used to describe a disease that occurs when cells undergo abnormal changes, leading to uncontrolled growth and division. While most cells in the body have specific functions and lifetimes, cancerous cells lack the normal mechanisms that regulate cell division and death. As a result, cancerous cells continue to grow and divide, depriving normal cells of oxygen and nutrients. They can form tumors and have the potential to spread to other parts of the body, damaging the immune system and disrupting normal bodily functions. Lung cancer specifically refers to the malignant growth of cells in the lung tissues, and it is a leading cause of cancer-related deaths worldwide. It's important to continue researching and developing methods for early detection and treatment of lung cancer, as it remains a significant health concern.[1] According to the survey of According to the World Health Organization (WHO), lung cancer was the second leading cause of death in 2015 and had dropped to the fifth rank by 2017. It is worth noting that smoking is the leading cause of lung cancer, accounting for approximately 85% of all cases. In recent years, there have been significant advancements in the development of Computer-Aided Diagnosis (CAD) systems for lung cancer. These systems utilize computer algorithms and image analysis techniques to assist radiologists and physicians in the detection and diagnosis of lung cancer. CAD systems can help improve accuracy and efficiency by highlighting potential abnormalities or nodules in medical imaging, such as CT scans. The early detection of lung cancer is crucial for preventing deaths and improving survival rates. Lung nodules are small masses of tissue that can be either cancerous (malignant) or non-cancerous (benign). Benign nodules are typically non-threatening and do not exhibit much growth, whereas malignant nodules grow rapidly and can spread to other parts of the body, posing a significant health risk. By utilizing advanced imaging techniques and computer-aided diagnosis systems, healthcare professionals can identify lung nodules at an early stage, enabling prompt treatment and better patient outcomes.For medical imaging so many different types of images are used but Computer Tomography (CT) scans are generally preferred because of less noise. Deep learning is proven to be the best method for medical imaging, feature extraction and classification of objects. The primary goal of our research is to diagnose the presence of lung cancer cells based on attributes, which are set of human symptoms, and information. The study explores the possibility of using an Artificial Neural Network model to detect the presence of a lung cancer in someone's body.

The purposes of this study are:

- To recognize some appropriate factors that cause lung cancer
- To model an Artificial Neural Network that can be used to detect the presence of lung cancer

Artificial neural networks (ANNs) are alike to our neural networks and offer a quite good technique, which solves the problem of classification and prediction[2]. An ANN is a mathematical model that is encouraged by the organization and functional feature of natural neural networks[3], Neural networks involve input and output layers, as well as (in most cases) hidden layers that transform the input into something so the output layer can use. When a neural network used for cancer detections, the ANN Model go through two levels, training and validation. First, the network is trained on a dataset. Then the weights of the connections between neurons are fixed so the network is validated to determine the classifications of a new dataset.

Contributions:

- ResNet, DenseNet, VGG, MobileNet and EfficientNet based CNN model developed to accurately extract information from CT scans.
- Transfer Learning is employed to gain high accuracy in less amount of time.
- Access to GPU based Detection system which was enabled with help of Google Colab

II. LITERATRE REVIEW

Jung, H., Kim, B., Lee, I. *et al.*[41] used two 3D deep convolutional neural networks (DCNNs) with shortcut connections and dense connections for the task of nodule classification. These networks are specifically designed to work with 3D data, allowing them to effectively capture the spatial information of lung nodules. The researchers found that the 3D DCNNs with shortcut connections and dense connections were able to capture general as well as distinctive features of lung nodules. The three-dimensional structure of the DCNNs proved suitable for extracting spherical-shaped features of nodules. To further improve performance, a checkpoint ensemble method was applied to the 3D DCNNs. This ensemble method involves training multiple instances of the network and combining their predictions to obtain a more robust classification outcome. The performance of the 3D DCNNs was evaluated using the LUNA16 dataset, which is a publicly available dataset commonly used for lung nodule classification research. The results showed that the proposed nodule classification method outperformed existing state-of-the-art methods for nodule classification. Overall, this research demonstrates the effectiveness of 3D DCNNs with shortcut connections, dense connections, and ensemble methods for accurate nodule classification, potentially contributing to improved diagnosis and treatment of lung nodules.

Nasrullah, N.; Sang, J.; Alam, M.S.; Mateen, M.; Cai, B.; Hu, H.[5] described a system for detecting and classifying malignant nodules in 3D lung CT images using deep learning techniques. The system consists of two main components: a 3D Faster R-CNN with CMixNet and a U-Net-like encoder-decoder. The 3D Faster R-CNN with CMixNet is used to detect the presence of nodules in lung CT images. Faster R-CNN is a popular object detection framework that utilizes a region proposal network to identify potential regions of interest. CMixNet is likely a customized or modified neural network architecture used in conjunction with Faster R-CNN for improved performance. Once nodules are detected, they are further analyzed using a 3D CMixNet with GBM (Gradient Boosting Machine) for nodule classification. GBM is a machine learning algorithm that combines multiple weak models to create a strong classifier. The CMixNet architecture is likely designed specifically for nodule classification and incorporates the features extracted from the detected nodules. The deep learning-based nodule classification results are then evaluated using various factors such as patient family history, age, smoking history, clinical biomarkers, nodule size, and location. These factors are likely used as additional features to enhance the classification accuracy and provide more comprehensive analysis of the detected nodules. Overall, this system combines deep learning techniques, including Faster R-CNN, CMixNet, and GBM, to detect and classify malignant nodules in 3D lung CT images. By considering multiple factors and utilizing advanced deep learning models, the system aims to improve the accuracy and efficiency of nodule classification, potentially assisting in the early diagnosis and treatment of lung cancer.

Yu Gu, Xiaoqi Lu, Lidong Yang, Baohua Zhang, Dahua Yu, Ying Zhao, Lixin Gao, Liang Wu, Tao Zhou[6] A 3D deep convolutional neural network (CNN)-based framework with a multi-scale nodule prediction strategy is proposed for automatic lung nodule detection in chest CT scans. This framework addresses the challenge of detecting lung nodules with significant variations in size within volumetric CT data. By adopting a 3D architecture, the CNN model is able to leverage the spatial information present in the CT scans, enhancing the accuracy of nodule detection. The multi-scale prediction strategy allows the network to capture nodules of different sizes by incorporating multiple scales during the detection process. This approach demonstrates the significance and effectiveness of the 3D multi-scale prediction strategy when dealing with lung nodules that exhibit large variations in size. The framework's ability to handle diverse nodule sizes improves the overall performance of the detection system, enabling it to accurately identify nodules in volumetric chest CT scans. By leveraging the power of deep learning and incorporating multi-scale predictions, this framework offers a promising solution for automatic lung nodule detection, contributing to early diagnosis and improved treatment outcomes in lung cancer.

S. Sasikala, M. Bharathi, and B. R. Sowmiya[7] proposed that CNN be used on CT scan images to detect and categorize lung cancer[8]. The researchers utilized MATLAB for their work and implemented a two-step training process for their proposed technology. In the first phase, they focused on extracting meaningful volumetric features from the input data, which likely involved preprocessing and feature engineering techniques specific to volumetric data. In the second phase, the extracted features were used for classification purposes, distinguishing between cancerous and non-cancerous cells. It is mentioned that their proposed technology achieved an impressive 96 percent accuracy in this task. By accurately differentiating between cancerous and non-cancerous cells, their technology shows great potential for aiding in the diagnosis and management of cancer, potentially leading to improved patient outcomes.

III. DATASET

The dataset Used is chest CT scan images and is available on Kaggle. The dataset contains a total of 1064 CT scan images, which have been labelled as either containing lung nodules or being normal scans. The images are in JPG format, which is a standard format used in medical imaging. The Images in Dataset are separated into three categories- Test, Train and Valid. The test, Train and Valid Categories are again separated into four categories- Images of Normal CT Images which do not contain tumor, Adenocarcinoma, Large Cell Carcinoma and Squamous Cell Carcinoma which contain images inside.

The Kaggle lung cancer dataset contains 2D CT (Computed Tomography) scans of the lungs. CT scans use X-rays to produce detailed images of the body's internal structures, including the lungs. CT scans are commonly used to diagnose and monitor lung cancer, as they can detect small nodules and other abnormalities that may be indicative of the disease.

CT scans are particularly useful for detecting lung cancer due to their ability to produce high-resolution images of the lungs.

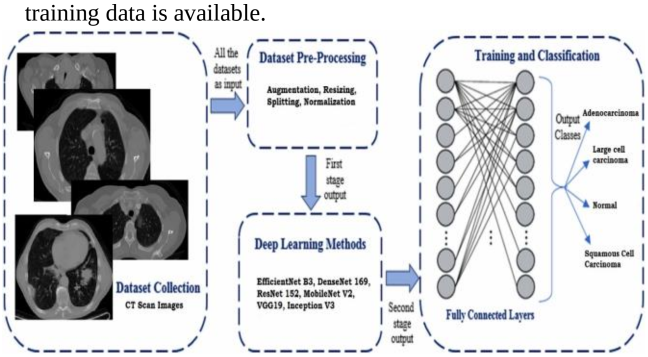
The dataset is intended to be used for the development of algorithms that can accurately detect and classify lung nodules in chest CT scans. Lung nodules are small, round growths in the lung tissue that can be indicative of lung cancer. Detecting and diagnosing lung nodules early is important for improving patient outcomes and survival rates.

IV. DESCRIPTION

Transfer learning is a technique used in machine learning that allows the knowledge learned from one task to be applied to another related task. In the case of medical image analysis, transfer learning using convolutional neural networks (CNNs) can be applied to improve the accuracy of lung cancer detection.

CNNs are a type of deep learning model that have been shown to be effective in image recognition tasks. Transfer learning using CNNs involves reusing pre-trained CNN models that have been trained on large image datasets, such as ImageNet, to extract features from new medical images.

To apply transfer learning in lung cancer detection, one approach is to use a pre-trained CNN model, such as VGG, ResNet, or Inception, that has been trained on a large dataset of natural images. The pre-trained model can be fine-tuned on a smaller dataset of lung CT images by replacing the output layer of the pre-trained model with a new layer that has the number of output classes equal to the number of cancer types to be detected. The fine-tuning process involves training the model on the smaller dataset of lung CT images, while keeping the pre-trained weights fixed. This allows the model to learn to recognize the specific features relevant to lung cancer detection, while leveraging the general features learned from the pre-trained CNN model. In addition, transfer learning using CNNs can also help to overcome the problem of limited training data, which is a common challenge in medical image analysis. By reusing pre-trained CNN models, the need for large annotated datasets is reduced, which can save time and resources. Overall, transfer learning using CNNs is a promising approach for improving the accuracy of lung cancer detection, especially in cases where limited training data is available.



By using data augmentation, the model is exposed to a larger and more diverse set of training examples, which can improve its ability to generalize to new, unseen data. In addition, data augmentation can also help to prevent overfitting by introducing randomness into the training process and making it more difficult for the model to simply memorize the training examples.

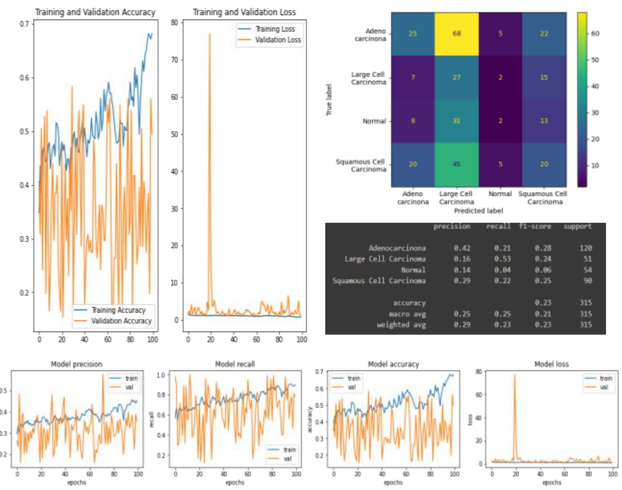
There are many pre-trained models available for image classification. Here are some models used in the system:

1. VGG19
2. ResNet152
3. MobileNetV2
4. EfficientNetB3
5. DenseNet169

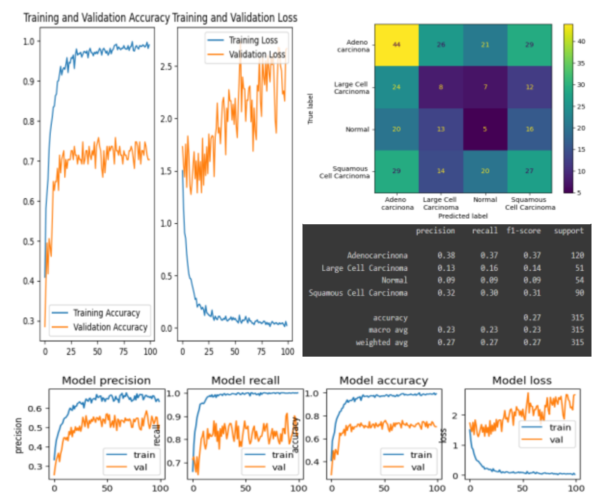
The Model uses Gaussian Classifier which helps in creating a robust environment, increase the speed of the process, automatically handling the missing data in the dataset and has the clearest interpretation that is useful to understand underlying structure of the data.

Here are the models along with their accuracy used in system:

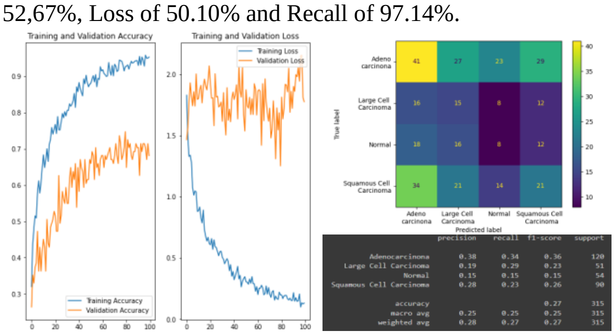
1. VGG- The VGG19 model along with the current dataset results in an accuracy of 49.52% , Precision of 36.79%, Loss of 121.17% and Recall of 76.51%.

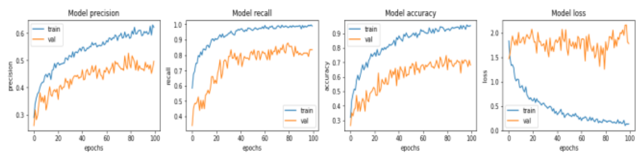


2. ResNet- The ResNet152 model along with the current dataset results in an accuracy of 96.19% , Precision of 58.21%, Loss of 24.72% and Recall of 99.05%.

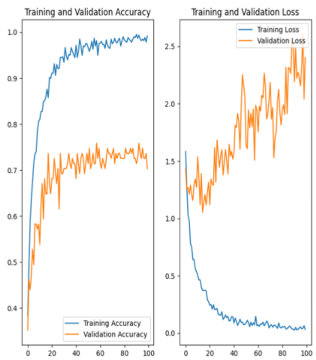


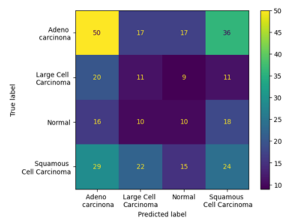
3. MobileNet- The MobileNetV2 model along with the current dataset results in an accuracy of 86.35% , Precision of 52,67%, Loss of 50.10% and Recall of 97.14%.



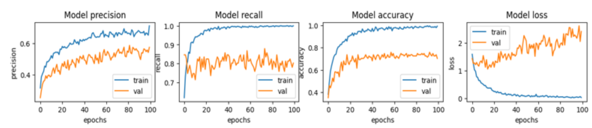


4. EfficientNet- The EfficientNetB3 model along with the current dataset results in an accuracy of 90.16% , Precision of 57.57%, Loss of 38.42% and Recall of 97.78 %.

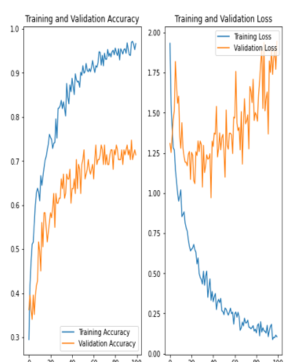


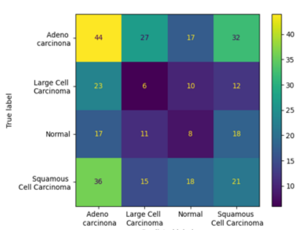


	precision	recall	f1-score	support
Adenocarcinona	0.43	0.42	0.43	120
Large Cell Carcinoma	0.18	0.22	0.20	51
Normal	0.20	0.19	0.19	54
Squamous Cell Carcinoma	0.27	0.27	0.27	90
accuracy			0.30	315
macro avg	0.27	0.27	0.27	315
weighted avg	0.31	0.30	0.30	315

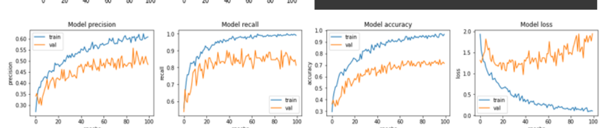


5. DenseNet- The DenseNet169 model along with the current dataset results in an accuracy of 93.65% , Precision of 60/59%, Loss of 30.11% and Recall of 98.10%.

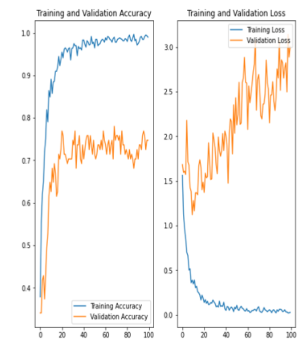


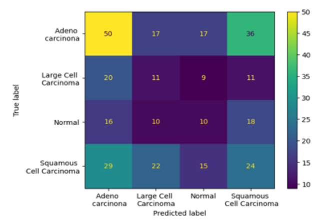


	precision	recall	f1-score	support
Adenocarcinona	0.37	0.37	0.37	120
Large Cell Carcinoma	0.10	0.12	0.11	51
Normal	0.15	0.15	0.15	54
Squamous Cell Carcinoma	0.25	0.23	0.24	90
accuracy			0.25	315
macro avg	0.22	0.22	0.22	315
weighted avg	0.25	0.25	0.25	315



Further the used dataset is equalized for better enhancement of the images in the dataset and highlight the underlying features of the images by increasing the contrast.
Since the ResNet obtained the highest accuracy in the above process we will use ResNet for further processing.
ResNet Model us used with Equalized dataset and removing Gaussian for further analysis-
The Model with above mentioned characteristics Obtains an accuracy of 90.07%, Precision of 53.51%, Loss of 53.25% and Recall of 96.03%.





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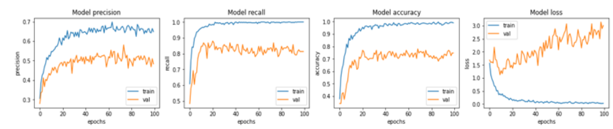
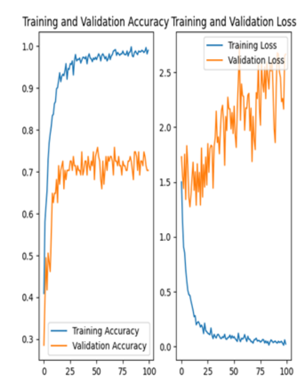
accuracy			0.30	315
macro avg	0.27	0.27	0.27	315
weighted avg	0.31	0.30	0.30	315

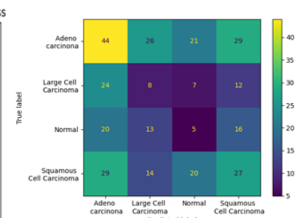


Since the accuracy decreased by removing gaussian and using and using equalized dataset we next tried to change the optimizer from Adam to SGD and ADAgrad.
The Model with SGD as optimizer obtained an accuracy of 54.60%, Precision of 43.41%, Loss of 128.72% and Recall of 71.11%
The Model ADAgrad as optimizer obtained an accuracy of 53.97%, Precision of 41.00%, Loss of 114.45% and Recall of 83.17%

V. RESULTS

From the Above analysis we can conclude that the Model using ResNet Architecture along with Adam as Optimizer and Original Dataset has obtained the highest accuracy and can be used for further processing and Evaluation.
The Confusion Matrix and further details are tagged along the image below.





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accuracy			0.27	315
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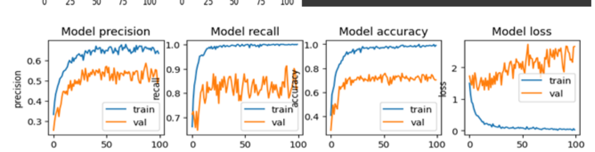
weighted avg	0.27	0.27	0.27	315



VI. CONCLUSION

The data analysis conducted on the proposed system for lung cancer detection demonstrated better performance compared to many existing programs in terms of accuracy, sensitivity, and specificity. This indicates that the system has the ability to accurately identify cases of lung cancer while minimizing false positives and false negatives. This indicates the system's stability and robustness.

By conducting a controlled variable experiment, the researchers were able to assess the individual effects of these factors. The results revealed that these factors significantly contribute to the system's overall performance in lung cancer detection.

The findings of this study not only highlight the effectiveness of the proposed system but also provide a foundation for future research. The study suggests that further investigation into fine-tuning ResNet and optimizing the current model could potentially lead to even higher accuracy in lung cancer detection. This opens up avenues for future studies to explore these factors in more depth and potentially enhance the system's performance further.

VII. REFRENCES

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